

Applications of Artificial Intelligence

Unit 8 Consolidate Task

Activity 1: LLM Exploration

Choose one of the following publicly available LLMs (ChatGPT, Claude, DeepSeek, Qwen, Llama, Gemini, or Mistral) and explore its capabilities.

For example:

Ask it to explain a complex concept from your field of study

Request a creative short story based on two random words

Ask it to translate a paragraph into another language

Challenge it with a logical reasoning problem

Ask it to solve a practical task (such as ones you might encounter in your daily life or study)

Document your findings. What did it do well? Where did it struggle? Experiment with different LLMs. Are there differences?

First LLM: Claude

1. **Writing a creative short story (Prompt: Write a 200-word creative short story using the words: "pencil" and "jogging")**

Claude's response is a creative but inconclusive story. It also overuses the em dash punctuation mark (—), which is a well-known characteristic of LLM-generated content (Stanciu, 2025).

2. **Explaining a complex concept (Prompt: Explain the Modigliani–Miller theorem)**

Claude returns a well-reasoned response that correctly explains the theorem.

3. **Solving a practical task in accounting (Inventory valuation where budgeted production differed from actual production, applying IFRS accounting standards).**

Claude returns the correct answer in a reasoned and well-structured format. It uses tables and clear wording to make the working easy to follow.

Evaluation: Claude is good for computational tasks, and for explaining complex concepts. However, other AI models return better results for creative tasks like story writing.

Second LLM: Mistral (Le Chat)

Public link to chat: <https://chat.mistral.ai/chat/b95e50e1-6175-44d3-af86-bc538d8a3e8d>

1. **Writing a creative short story (Prompt: Write a 200-word creative short story using the words: "pencil" and "jogging")**

Le Chat returns a more creative story than Claude did with the same prompt. However, it still over-uses em-dashes and adjectives.

2. **Explaining a complex concept (Prompt: Explain the Modigliani–Miller theorem)**

The response captures the general theme of the theorem but makes errors about its core propositions.

Le Chat's response is also less structured than Claude's response to the same question.

3. **Solving a practical task in accounting (Inventory valuation where budgeted production differed from actual production applying IFRS accounting standards).**

Le Chat arrives at the correct answer but has a less reasoned and concise approach than Claude. Notably, it calculates the total production cost, which is not a necessary or relevant step in the process.

Evaluation: Le Chat is a good tool for research but is less refined and has less concise responses (at least in its default settings).

Third LLM: Web LLM

1. **Writing a creative short story (Prompt: Write a 200-word creative short story using the words: "pencil" and "jogging")**

Web LLM's output is a genuinely engaging story that does incorporate the words given as key concepts in the story. Also, it doesn't overuse adjectives or include em dashes, as many LLMs would.

2. **Explaining a complex concept (Prompt: Explain the Modigliani–Miller theorem)**

Web LLM gets the explanation wrong, assuming the Modigliani-Miller theorem to be an algebraic geometry concept, when it is really a corporate finance theorem. The LLM also returns "hallucinated" details about the theorem and its propounders.

3. **Solving a practical task in accounting (Inventory valuation where budgeted production differed from actual production applying IFRS accounting standards).**

Web LLM gets the calculation wrong. It calculates the cost of production but applies an overtly wrong formula for calculating the inventory value (it calculates this as variable cost minus variable cost, which of course returns a value of 0).

Evaluation: Web LLM is a good tool for creative tasks (like story writing and idea brainstorming), but not for research, computational, or reasoning tasks.

Activity 2: Prompt Engineering

Start with a basic, general prompt, such as "Tell me about climate change"

Gradually refine your prompt through 3-4 iterations to get more specific, detailed, and useful information

For each iteration, note:

Your modified prompt

How the response changed

What techniques made the most difference

Identify 3 key principles of effective prompt design based on your experience

Prompt Engineering Scenario: The Digital Divide

Scenario: I am a secondary school teacher preparing a lesson on the digital divide for a Year 8 class. I am using an LLM to brainstorm the causes and effects of the digital divide in the UK, and to word my lesson content in an engaging manner for the target age range.

LLM Used: Claude

Public link to LLM chat: <https://claude.ai/share/d6dac83f-83a9-4b1d-bb6e-95a237f1c1b8>

Prompt 1: Tell me about the digital divide

Notes: This prompt provides a general and vague answer that simply defines the issue and details some effects.

Prompt 2: Explain the causes and effects of the digital divide in the UK

Notes: The answer to this prompt goes into more detail, but mainly cites statistics and figures from external websites, rather than analysing or discussing the issue as required in the context.

Prompt 3: Explain the causes and effects of the digital divide in the UK in a manner suitable for an older child.

Notes: This answer includes multiple causes and effects, and is framed in an age-appropriate manner, in more accessible language, and with simpler vocabulary. Adding details of the audience makes some difference and helps the LLM to respond more appropriately to the scenario.

Prompt 4: You are a PHSE (Personal, Social, Health and Economic Education) teacher in an English secondary school. In 400 to 500 words, discuss the social and economic causes and effects the digital divide in language suitable for 12 and 13-year-olds in a mainstream urban secondary school. Give three points each for causes and effects, and ensure the language is age-appropriate for the audience.

Notes: This is clearly the best and most suitable answer. Giving specific parameters and context allows the LLM to tailor its response and wording to the desired audience. For example, by clarifying the lesson is for students in an urban school, the LLM gives an answer tailored to students who likely have access to digital resources. The language is also exciting and engaging.

Prompt 4 was the most specific and made the most difference.

Principles of Effective Prompt Design:

1. **Be Specific:** Specify output length and format.
2. **Give context:** Provide some context on the scenario in which the output will be used and specify the tone in which the output should be delivered.
3. **Assign a role:** Assigning a role to the LLM helps to tailor the output to be suitable for the situation and audience.

Activity 3: LLM Limitations

1. Deliberately try to expose the limitations of an LLM by:

Asking for very recent information (beyond its training data)

Requesting complex mathematical calculations

Asking for personal opinions on controversial topics

Probing for factual knowledge in a niche area

Document where the LLM acknowledges its limitations versus where it "hallucinates" plausible but incorrect information

LLM Used: Web LLM Chat

Prompt: Mathematics question on arithmetic and geometric progression.

Result/Comment:

The LLM solves the question correctly for the first few steps. However, it makes a mistake mid-way into the calculation and returns a wrong final result.

Prompt: In UK corporation tax reducing balance capital allowances, what is the current main pool rate?

Result/Comment

This requires knowledge of very recent information because the related tax law has recently been changed at the time of completing this assessment (which is 30th April 2026). The current correct answer is 14% (It was 18% until 1st April 2026).

Web LLM returns a “hallucinated” answer that is unrelated and with the wrong rate.

Claude, (which is generally considered well-suited for finance-related and computational tasks, (Jiang, 2025)) returns 18%, reflecting an understanding of the question, but not updated knowledge.

Prompt: How many restaurants can be found on Great Horton Road in Bradford, England?

Result/Comment:

Web LLM recognises its limitations here by stating that it cannot find the requested information, but suggests good possible sources of information, such as Google Maps and local directories.

References

Jiang, Q., Gao, Z., & Karniadakis, G. E. (2025). DeepSeek vs. ChatGPT vs. Claude: A comparative study for scientific computing and scientific machine learning tasks.

Theoretical & Applied Mechanics Letters, 15(3), Article 100583.

<https://doi.org/10.1016/j.taml.2025.100583>

Stanciu, A. (2025). Why did LLMs steal our em-dashes? *McGill University Office for Science and Society*. <https://www.mcgill.ca/oss/article/critical-thinking-student-contributors-technology/why-did-llms-steal-our-em-dashes>